

Time, Derived: The Commitment-Flow Time of D_{IV}^5 and Its Causal Action

Time as the flow of the conformal Hamiltonian on a bounded
symmetric domain — emergent, not assumed; with a spectral action
that is a causal action, and a degree-2 cover that is exactly
spin-statistics

Casey Koons, with Lyra, Keeper, Elie, Grace, Cal A. Brate (CI co-authors / referee)

2026-09-30 (v1.5; v1.4 of 2026-09-27 is the prior record, unchanged;
v1.3 of 2026-08-17 is the hashed GO file, unchanged)

Time, Derived, in D_{IV}^5

Abstract

We show that in Bubble Spacetime Theory time is not a background parameter but the flow of a single operator — the linear conformal Hamiltonian J on the substrate Hilbert space $H^2(D_{IV}^5)$. Time is emergent: it is the flow parameter of the commitment semigroup, its arrow is the positivity of J 's spectrum, and the generator itself is forced by the geometry (the linear $SO(2)$ -center charge, not the Casimir). The one operator has two faces — a real-time tick and an imaginary-time circle — the two boundary values (Wick pair) of one holomorphic semigroup. The paper's one substantive result is that time is derived — with an explicit generator, arrow, and Wick structure; everything else we label honestly as a consistency or a scaffold. An earlier draft made two over-identifications that we correct here, for *independent* reasons: electric charge is not the time-generator (it is not even a function of J — the muon and neutrino share a conformal weight but differ in charge), and Connes' spectral action is not $\text{Tr } f(J)$ (it needs a Dirac operator whose square is Laplace-type, encoding curvature, which the positive J cannot supply). Accordingly the causal-order $\leftrightarrow$ spectral-action connection (Section 11) is marked Scaffold, now with its real operator — the built, sign-indefinite Kostant cubic Dirac D , whose D^2 Laplace-type heat trace is the F60/F63 Sakharov induced gravity — leaving the causal-order *weighting* as the open piece.

The imaginary-time circle is represented on a degree-2 cover, and we are careful about what that is and is not. The condition is referee-checkable ($T = 2\pi/\text{gcd}$ of the charges within a superselection sector; $4\pi \cdot \frac{1}{2} \text{gcd} = \frac{1}{2}$), and the parity content is an

exact identity: $\exp(2\pi i J) = (-1)^{\#\text{Rac}}$, the parity of a state's scalar-singleton (Rac) constituent count. On physical particles (two-singleton composites) this equals $(-1)^F$ in every dimension — it is exactly fermion parity: fermions ride the cover (720°), bosons do not. The degree-2 cover is therefore the geometric restatement of spin-statistics, not a novel prediction — we retract an earlier draft's claim that it is “forced by n_C being odd” (the dimension-dependence cancels between constituents; what the odd dimension decides is only which *constituent* carries the half-odd weight, and constituents are not particles). We report it as a consistency: the theory reproduces spin-statistics, and the gravitational sector rides the single cover (the graviton is even- $\#\text{Rac}$). We state plainly what we do not answer: the tick's numerical value, and the weighting in the spectral $\leftrightarrow$ causal action equality. A paper on time earns a referee for one reason — time is derived, and the paper says what it does not know.

1. Time is emergent [Derived — Tier-0]

The substrate Hilbert space is $H^2(D_{IV}^5)$, the Hardy space (the unitary highest-weight module at $\nu = n_C/2 = 5/2$) of the bounded symmetric domain $D_{IV}^5 = SO_0(5,2)/[SO(5)\times SO(2)]$ (forced as the unique automorphism-invariant Born space, T754). On it, the commitment dynamics is a one-parameter semigroup

$$\rho_{\text{commit}}(\tau) = \exp(-\tau J/\hbar).$$

Time is the flow parameter τ — not an external axis, but a reading of how far the commitment semigroup has run. “When” is a measure of accumulated commitment; there is no clock behind the clock.

Physical time is flat and one-way — say this at the outset, to forestall the natural misreading. The *physical* flow is the real-time semigroup $\exp(-\tau J)$: a one-directional half-line, with no recurrence — time does not loop back on itself. The “circle” that appears later (Section 5, Section 8) is the imaginary-time (Wick) face of the same generator and, physically, a selection rule on admissible states — a property of the analytic continuation and the state space, *not* a claim that physical time is periodic. Whenever this paper says “circle” or “double cover,” it means the Wick face / the selection rule; physical time is the arrow, full stop. (We list this explicitly in what the paper does not claim, Section 12.)

2. The generator is the linear conformal Hamiltonian J, not the Casimir [Structure-Derived]

The generator is the conformal Hamiltonian J — the $SO(2)$ -center weight of $K = SO(5)\times SO(2)$, with a half-integer spectrum. Its ground weight on H^2 is $5/2 = 3/2 + 1$: the minimal representation's (Rac's) $3/2$ plus one unit of the odd clock (Theorem B, $H^2 = \text{Rac} \otimes \text{odd clock}$). The singleton weights $\{3/2, 2\}$ generate the multi-singleton Fock space, where the parity law of Sections 5 and 7 lives. (*J is the generator of the flow exactly in the limit $\Lambda \rightarrow 0$. With $\Lambda > 0$ the cosmological curvature fixes*

a timelike vector in J 's own plane, so the surviving symmetry is de Sitter $SO(4,1)$ and the conserved time is a de Sitter boost. J then generates the flow to order H/E , where E is the energy of the process. J 's spectrum on H^2 is a property of the representation and is unchanged; the arrow the paper reads from it — the flow running one way — holds where J generates the flow, i.e. to order H/E , since the exact de Sitter time is a boost with two-sided spectrum.) Its eigenvalue on a state is the conformal weight E itself, and it is this half-integer ground weight that makes $\exp(2\pi i J) = -1$ (Sections 5, 7) — the double cover rests on the half-integer eigenvalue, never on a step size. (A note on the word “charge”: where this paper says “charge” — the gcd condition of Section 5, the #Rac counting of Section 7 — it means this $SO(2)$ -center weight, the conformal energy label. That is a different quantity from the electric charge Q of Section 3a, which lives on the $SO(5)$ factor; the two are kept strictly apart.) It is emphatically not the quadratic Casimir $C_2(K)$: a Casimir is *central*, constant on each irreducible representation, so it labels *which particle* and cannot evolve anything within an irrep. Only the non-central, linear J generates motion — and its linearity is exactly what makes $\exp(-iJt/\hbar)$ the standard Schrödinger equation. We hold the distinction sharply: J is the time; the Casimir is the particle-label — two operators on one representation tower, distinct roles. (This corrects the earlier Tier-0 draft, which mis-cast the Casimir as the dynamical generator.)

3. The arrow is spectrum-positivity [Derived]

J is bounded below ($\text{spec } J \geq E_0 > 0$). Hence $\exp(-\tau J)$ is a contraction semigroup, defined only for $\tau \geq 0$: the flow runs one way. That positivity is the arrow of time — not an added postulate, but the statement that the energy operator has a ground state. Energy generates, time counts, and the elementary tick is $\hbar/\text{energy}$. (The tick's numerical value $\approx 10^{-120}$ s is Identified, not Derived — see Section 8.)

3a. Charge is internal, not temporal [Derived: charge is internal; conservation a corollary]

Could electric charge be a reading of the *same* circle as time (charge = where a state sits, time = the circle turning)? It cannot — decisively, by degeneracy: the muon and the neutrino are both $\text{Rac} \otimes \text{Di}$ at the same conformal weight $\Delta = 7/2$, yet carry $Q = -1$ and 0 , so electric charge is *not a function of the time-generator at all* ($Q \neq f(J)$ for any f). That one line settles it, with no appeal to sign or positivity. A second, independent (and weaker) reason points the same way: Q is bounded and sign-indefinite (antiparticles) while J is bounded below and unbounded, so “the same one-dimensional circle” — which would force $Q \propto J$ — is doubly excluded. These are independent facts, not one principle; degeneracy is the load-bearing one.

The derived content is that electric charge is *internal* — it lives in the internal sector of $K = SO(5) \times SO(2)$, not on the conformal $SO(2)$ -center that carries time. We are careful about *where* internally: the integer part is electroweak (on the $SO(5)/SO(4)$ structure), but the fractional $\{\pm 1/3, \pm 2/3\}$ cannot come from the $SO(5)$ torus (B_2 's weight lattice has no thirds, and $SU(3)$ is not a subgroup of $SO(5)$). The thirds are

the fingerprint of the $N_c = 3$ color structure — and here we are careful in a way that protects load-bearing geometry: the geometry forces the *number* 3 (the short-root multiplicity of D_{IV}^5 's root system — an intrinsic invariant), but it does not host the *group* $SU(3)$ as an isometry ($SU(3) \not\cong SO(5)$, for two independent reasons: no A_2 subsystem in B_2 , and $SU(3)$ has no real 5-dimensional irrep). The centers are also unrelated: $SO(5)$ has *trivial* center, its double cover $Spin(5) = Sp(2)$ carries the $\mathbb{Z}_2$ (the rank-2 / $Pin(2)$ structure), and neither is $SU(3)$'s $\mathbb{Z}_3$. $SU(3)$ is a genuine continuous gauge group (gluons are real), realized *on* the 3-dimensional multiplicity space, not as a spacetime symmetry; its $\mathbb{Z}_3$ center is what appears in the charge grading (the thirds) and in confinement (only $\mathbb{Z}_3$ -neutral states reach the boundary). So the honest statement is “charge is internal, two-source,” not “charge sits on the $SO(5)$ torus.” Because the charge sector and the time-circle are different (commuting) factors, $[Q, J] = 0$, so charge is conserved under time evolution — but we state this as a consistency corollary of charge-being-internal, *not* as independent evidence: once charge is off the time-circle, the commutator vanishes by construction.

(Casey's electric $\leftrightarrow$ magnetic intuition has its correct home elsewhere: E and B unify in $F_{\mu\nu}$ on the conformal $SO(4,2)$ boundary, where a boost rotates one into the other — a genuine pairing, distinct from the charge $\leftrightarrow$ time one that does not unify.)

4. The two faces — the Wick pair [Derived; standard theorem, cited not claimed]

Because J is self-adjoint and bounded below, $\exp(-zJ)$ is a holomorphic semigroup on the closed right half-plane $\text{Re } z \geq 0$ (Stone / Hille–Yosida). Its two boundary faces are: - Real-time face $\exp(-\tau J)$ ($z = \tau$): the irreversible tick — a one-directional half-line. - Imaginary-time face $\exp(i\theta J)$ ($z = i\theta$): the reversible circle. These are the Euclidean and Lorentzian faces of the *one* generator, tied by the Wick rotation $\tau \leftrightarrow i\theta$. We stress that the existence of the two faces is a standard functional-analytic fact for any bounded-below generator; what is specific to BST is that the generator J is *forced by the geometry*, not that time has two faces.

5. The circle closes on a degree-2 cover [Derived-as-condition; the odd dimension decides which constituent is half-odd]

The imaginary-time circle's minimal period is fixed by the charge spectrum. The referee-checkable statement (which cannot be satisfied vacuously) is:

$$T = \frac{2\pi}{\text{gcd}\{\text{charges}\}} \text{ within a superselection sector, } T = 4\pi \iff \text{gcd}\{\text{charges}\} = \frac{1}{2}.$$

(The fork “is the double cover projective (charge differences only) or absolute (the charges themselves)?” is resolved absolute — Section 6; the closure is set by the charges, and the degree of the cover is the *denominator* of that gcd.) The half-integer step is supplied by the Rac scalar (conformal weight $E = 3/2$); the Di spinor ($E = 2$) is integer. Because $d = n_C = 5$ is odd, the scalar weight $(d-2)/2 = 3/2$ is the

half-integer, while the spinor weight $(d-1)/2 = 2$ is not. What the odd dimension decides is only *which constituent* carries the half-odd weight — here the scalar; in even dimension the roles swap (the spinor half-odd), but the cover survives on the same *physical* particles (Section 7). It does not decide whether the cover exists. The two weights are primary-sourced in two independent legs of Fernando–Günaydin (arXiv:1409.2185): §7 eq 7.1 (scalar minrep, $E = 3/2$) and §9 Table 2 (spinor, $E = 2$).

A retraction we keep visible (the method self-correcting). An earlier version of this result claimed “matter is spinorial from the shape of time.” That was wrong — the energy assignments were swapped. On the correct weights the story *inverts*: in odd dimension the scalar forces the double cover, and the spinor tower would close at 2π on its own. “Spinors are 4π -periodic” is a triviality and, for the *time* circle, here false. We keep the correction in print because it is evidence the framework is falsifiable against its own prettiness.

Terminology. This paper concerns time; the Higgs enters only as one boson among many (it rides the 2π single cover). Its substrate identity is now settled (Section 7) — a Lorentz scalar carrying an internal (1,0) vector — so we may name it, but nothing in the time results depends on that identification.

6. The fork resolved as a representation fact — but its observability is a separate, open question [rep-fact Banked; observability Open — D5]

Does a single state genuinely “ride” the 4π ? The representation fact is settled algebraically: on a half-odd-weight state $\exp(2\pi i J) = -1$ (Section 7), so the 2π flow is not the identity operator on the Hilbert space — the time circle is represented faithfully only on its degree-2 cover ($T = 2\pi/\gcd\{\text{charges}\}$, $4\pi \nmid \gcd = 1/2$). This is a statement about the operator, not a measurement. (We deliberately do *not* phrase it as an interference experiment: superposing an “evolved” and an “un-evolved” copy would require a clock external to the only time there is, which the time generator does not admit — the D5 obstruction of the next paragraph. Stating the rep-fact algebraically keeps Sections 6 and 12 consistent.)

And the 4π is a selection rule, not something you measure (D5). It is not observable by interferometry or spectroscopy: a two-arm interferometer measures phase *differences* — the projective quantity, period 2π — and the absolute charge that distinguishes 2π from 4π is exactly what it cannot see. Nor can one “hold an un-rotated copy” as a reference: a record stores the *outcome*, not the phase (the point is Keeper’s, from the commitment-as-record structure), and time evolution is not externally applicable to a single arm the way a magnetic field is. This is not a demotion (and not an over-claim of unphysicality: the absolute weight still *gravitates*, coupling to $T_{\mu\nu}$ — the claim is specifically about *local phase* observables). Its honest content is a selection rule: the degree-2 cover fixes which cover of the state space the theory represents, hence which states are admissible — which states exist, not a fringe. We do not claim the 4π is locally observable (Section 12).

7. The parity content — the degree-2 cover is exactly fermion parity [Derived, dimension-independent; a spin-statistics consistency, not a novel prediction]

J's eigenvalue is the conformal weight E *itself* — a half-integer, ground $E_0 = 3/2$ on the Rac and $5/2$ on H^2 (Section 2, not the electric charge) — so a 2π turn acts by $\exp(2\pi iJ) = \exp(2\pi iE)$. The singleton weights are forced by the Faraut-Korányi multiplicity $a = n_C - 2 = 3$: the scalar (Rac) at $E = 3/2$, the spinor (Di) at $E = 2$. Every conformal weight is $E = (3/2) \cdot \#Rac + 2 \cdot \#Di + n$ (n integer), and since $\exp(2\pi iE) = (-1)^{\{2E\}}$ with $2E \equiv \#Rac \pmod{2}$,

$$\boxed{\exp(2\pi iJ) = (-1)^{\#Rac}} \quad (\text{parity of the number of Rac scalar-singleton constituents}).$$

This is an exact identity on the weights. Its content on physical particles is sharper — and *dimension-independent*. A physical particle is a two-singleton composite (a premise; open per K1653), so $\#Rac + \#Di = 2$ is even, whence $\#Rac$ and $\#Di$ share parity; each Di carries half-integer spin, so $\#Di$ -parity is spin-parity, and

$$\exp(2\pi iJ) = (-1)^{\#Rac} = (-1)^{2\text{spin}} = (-1)^F.$$

The degree-2 cover of time is exactly fermion parity — fermions ride it (720°), bosons do not — and this holds in *every* dimension: the d -dependence cancels between the constituents, $(-1)^{\{d-2\}}(-1)^{\{d-1\}} = -1$ always. $\#Rac \pmod{2} = (-1)^F$ is a conserved $\mathbb{Z}_2$ selection rule (fermion parity). This identity holds on two-singleton composites. It is not an identity on the substrate H^2 : there $\exp(2\pi iJ) = -1$ on every state (all weights in $5/2 + \mathbb{Z}$) while every K-type has integer $SO(5)$ spin. H^2 is not a singleton composite (Theorem A), so the two statements do not collide; on H^2 the clock grading counts H^2 quanta mod 2, not fermions.

We retract the “forced by n_C odd” reading (a fifth self-correction of “the double cover is special”). An earlier draft called the cover “forced because $n_C = 5$ is odd — the paper’s core result.” That was the *constituent* statement kept after the composite reading replaced it: the parity is only odd *for the bare Rac constituent*, and constituents are not particles; on particles the phase is $(-1)^F$ in all dimensions. What the odd dimension decides is merely which constituent carries the half-odd weight (Section 5). So the degree-2 cover is the geometric restatement of spin-statistics — a consistency the theory must and does satisfy — not a novel BST prediction. We report it as such. (Cal’s honest core sentence: *time’s circle is represented on a degree-2 cover, and the cover is exactly fermion parity; the odd dimension decides only which constituent is half-odd.*)

We accordingly retract the earlier “only the Higgs rides 4π ”: it required treating a *bare singleton* as a particle, which fails the identity filter — a bare Rac is a spin-0 object with $\#Rac = 1$, i.e. fermionic parity on a scalar, so it is neither a boson nor a spin- $1/2$ particle. (The same filter closes an old corpus contradiction: “Rac = Higgs” fails on parity, “Rac = muon” fails on spin — both identify an SM particle with a bare singleton, and the composite reading deletes both at once.) The photon is the

$\Delta = 4 \text{ Rac} \otimes \text{Rac}$ composite ($\# \text{Rac} = 2 \rightarrow 2\pi$), consistent with the measured 360° return of photon polarization.

The Higgs identity (a check, not a load-bearing claim). Spin-statistics does not depend on it: the Higgs is a boson on the 2π cover. From the primary source (Fernando–Günaydin), the Rac minrep is a massless conformal scalar whose own level-1 K-type is the internal (1,0) vector; that level-1 mode is a state of *one* Rac (weight 5/2, clock–spin mismatched) and is not the Higgs. The Higgs is read as the $\text{Rac} \otimes \text{Rac}$ scalar ($\Delta = 3$; $\# \text{Rac} = 2$, on the 2π cover), whose (1,0) K-type at weight 4 carries the (2,2) bi-doublet direction of the boundary condensate (F603; K1197). *(v1.4 called the single-Rac level-1 mode and the $\text{Rac} \otimes \text{Rac}$ scalar “two aspects of one object”; they are different states of different modules — distinct irreducibles with different lowest weights, so no G-map joins them (Schur) — and that sentence is withdrawn. Lyra R22; Cal Section 1020; Grace R182.)* This does not touch the derived time content.

Scope. The identity uses the free-field singleton weights $\{3/2, 2\}$; it is not protected against anomalous dimensions. (An “antipodal map” for a deeper mechanism faces one obstruction found twice — $\exp(i\pi J) = -i$ is a fourth root of unity, no involution produces it, and the same obstruction recurs via the Pin^- structure — and it suffices: the mechanism is the forced weight, no deeper object.)

8. The geometry of the path: one circle, twice (a Möbius bundle) — kept separate from the Wick pair [Derived geometry; meaning open]

The time-generator is the one $U(1)$ center of K — not an $SU(2)$, not two circles (the $SU(2)$ ’s live in $SO(5)$ and carry space/spin, not time). The relevant cover is the degree-2 cover of that circle — *not* “ $\text{Spin}(2)$ ”: $\pi_1(SO(2)) = \mathbb{Z}$, so covers exist at *every* degree, and the degree we get is the denominator of $\gcd\{\text{charges}\}$ (a spectrum with $\gcd \frac{1}{3}$ would close at 6π). We avoid “ $\text{Spin}(2)$ ” as a *group name* — $\pi_1(SO(2)) = \mathbb{Z}$, so covers exist at every degree, and this is a degree-2 cover of a $U(1)$, not the spin group $\text{Spin}(2)$. (Its physical *content* is nonetheless fermion parity, Section 7 — the cover is a property of the charge lattice that happens to coincide with the spin grading on physical particles.) The degree-2 cover of a circle is a circle “twice as long,” so the path is one circle traversed twice — the *same* statement as Section 5 ($\gcd = \frac{1}{2}$ degree 2). A half-integer-charged state is a section of the corresponding real line bundle (a Möbius edge) over that circle — a frame flips sign at 2π and returns at 4π .

We keep this strictly separate from the two faces of Section 4. The two faces (real tick / imaginary circle) are the *Wick pair* — the complexification of the time parameter, with the faces as the real and imaginary axes of complex time z . The Möbius double cover is a winding property inside the imaginary axis alone. They are two different structures: one is real-vs-imaginary time, the other is how many times one loops the imaginary circle. In particular the picture “the two faces are the two sides of one Möbius strip” is doubly false — the structures are different, and a Möbius strip has *one* side. Nor is it “two directions of time”: real time is a half-line (one direction,

the arrow); only the imaginary circle is bidirectional.

9. What ticks for a photon? — substrate time vs proper time [participation answered; the photon’s cover-status open]

A massless particle (the photon; the $m_1 = 0$ neutrino) has a null worldline and hence zero proper time — it does not age. This does *not* place it outside substrate time: it participates through its energy, the tick being $\hbar/\text{energy}$ (Section 3). Proper-time-zero $\neq$ non-participation — two different clocks, and the substrate clock (J’s flow, energy-driven, universal) runs while the worldline clock (zero on the null cone) does not.

How a delocalized massless state localizes its commitment — resolved. A definite-energy massless state is delocalized, but its commitment is not a bulk position: it is a Szegő projection to a point on the celestial boundary (an S^4 direction plus a phase), not a location in the interior. This is exactly why a photon has no bulk position operator — its commitment lives on the boundary, not in the bulk.

The photon rides the single cover — it is a boson. The gauge boson is the $\Delta = 4$ $\text{Rac} \otimes \text{Rac}$ composite ($\#\text{Rac} = 2$, even $\rightarrow 2\pi$ by the $\#\text{Rac}$ law), consistent with the measured fact that photon polarization returns at 360° . (A bare vector singleton at $5/2$ is excluded: there is no vector singleton, and a $5/2$ spin-1 primary is non-unitary below Mack’s bound $\Delta \geq 4$ — the only $5/2$ object is the descendant $\partial_\mu(\text{Rac})$, not a particle.) So the photon rides 2π — an instance of the forced spin-statistics of Section 7 (a spin-1 boson turns 360°), not a contingent assignment. Its *participation* in substrate time (via energy) and its Szegő celestial localization, above, are unaffected and stand.

10. Spin-statistics — the degree-2 cover is fermion parity [Derived, dimension-independent; with its falsifiers]

On physical particles, the two-singleton composites of Section 7, the time double cover is spin-statistics: $\exp(2\pi i J) = (-1)^{\#\text{Rac}} = (-1)^{2 \cdot \text{spin}} = (-1)^F$ identically (Section 7), in every dimension. The cover lands on a state *iff* the state is a fermion — fermions turn 720° in time exactly as they do in space. $K = \text{SO}(5) \times \text{SO}(2)$ is a direct product, so the spatial-spin circle ($\text{Spin}(5)$) and the time-conformal circle (J) are *independent* generators, and their coincidence on the fermions is the content: the time double cover is fermion parity, a conserved $\mathbb{Z}_2$ selection rule. This is a *consistency* the theory reproduces, not a new prediction — and it is dimension-independent, so it is not “forced by n_C being odd” (that reading, retracted in Section 7, was the constituent-level statement mistakenly kept at particle level).

The falsifiers — what the result rests on. Two explicit, referee-checkable assumptions: (i) the spectrum is generated from $\{3/2, 2\}$ by addition ($\text{gcd} = 1/2$); and (ii) every populated sector lies in that ladder. Break either and the parity relation of Section 7 breaks with it. And as a caveat: all of it uses the free-field / singleton

weights and is not protected against anomalous dimensions. These are the statements a referee should attack; the degree-2 cover and the parity relation stand or fall on them together.

11. The causal order and a spectral action — a scaffold, honestly labeled [Scaffold]

Derived time carries a genuine causal structure: the commitment order on the Shilov boundary is a derived causal set (a locally finite partial order, with the order *forced* by the commit operator — the piece standard causal-set theory must posit). It is natural to ask whether BST’s own causal action over that order equals a spectral action on the geometry. We record this as a program, not a result — and correct an over-identification an earlier draft made.

What the object is *not*. The spectral action of induced gravity is not $\text{Tr } f(J)$. J is positive (it is the arrow, Section 3), so $\text{Tr } f(J)$ is a partition function — thermodynamics, not curvature. Connes’ spectral action is built from the Dirac operator D , which is *sign-indefinite* (it encodes chirality and the metric); it is D^2 — a Laplace-type operator — whose heat expansion yields the Einstein–Hilbert action. The reason is definite (and *not* a general “arrow principle” — Section 3a): the spectral action requires a Dirac operator whose square is Laplace-type, encoding the metric’s curvature, which the positive J simply cannot supply. And that Dirac operator is not hypothetical — BST has built it (the Kostant cubic Dirac on $\Lambda(\mathbb{C}^5) \otimes H^2(D_{IV}^5)$, *self-adjoint and credentialed*; a *genuinely* sign-indefinite* operator, unlike J). Its D^2 is Lichnerowicz Laplace-type (the domain Laplacian plus a curvature term), and from its Seeley–DeWitt expansion the Einstein–Hilbert density falls out with a definite (positive) sign on the bulk geometry (F60/F63) — a sign fixed by the bulk being *negatively* curved ($R < 0$, AdS-like), which the boundary sphere ($R > 0$) could not have shown. We state the tier honestly: this is a *density with a sign*, not “BST induces gravity with coefficient X .” The bulk is non-compact (infinite volume), so the total heat trace is not trace-class; only the *local* Seeley–DeWitt densities are finite (exactly Sakharov’s quantities). As for the *value* of Newton’s constant: it is one dimensional input — the boundary curvature radius ℓ_B — exactly as general relativity takes G as an input. BST supplies the dimensionless *structure* (the Einstein–Hilbert density with its definite sign, above); the single scale ℓ_B then sets G . This is the same bargain GR makes, not a shortfall: the structure is derived, the one scale is taken. (We claim no specific value for G here; whether anything fixes ℓ_B in absolute terms — turning that input into a prediction — is a separate open question.) So Connes’ spectral action is $\text{Tr } f(D/\Lambda)$ with this D , not J ; the Einstein–Hilbert density and its sign are on the right (bulk) manifold, and the finite $1/G$ is open.

Why the base camp is not evidence. One can write $\int P(x,y)P(y,x) d\mu(y) = P(x,x)$ (the Bergman reproducing property, a two-point→one-point collapse) with $\text{Tr } P =$ the state-space dimension — but this idempotency is construction-guaranteed (it holds identically whether or not the two actions are related), so it is not evidence. And the content that *would* matter is the weighting: in a causal-set action the weights

are the entire content (Benincasa–Dowker’s alternating coefficients are what make it converge to the curvature); an unweighted pair-sum is not a causal action. That weighting is not derived. So this is a scaffold — but a better-founded one: the operator is built (the Kostant cubic Dirac D), its spectral action $\text{Tr } f(D)$ connects to the banked induced gravity (F60/F63), and what remains open is the weighting — whether BST’s derived commitment-order supplies pair-weights matching that spectral cutoff. [*Scaffold — object built and named; weighting open.*]

11a. Gravity rides the single cover; the induced-gravity supertrace [framework-tier note]

The double cover is a fermionic phenomenon, and we state one boundary of it plainly to forestall over-reading. The graviton is $\Delta(T_{\mu\nu}) = d = n_C = 5$ (two routes agree — CFT stress-tensor conservation at spin 2, and the Flato–Fronsdal $\text{Rac} \otimes \text{Rac}$ $\ell=2$ content give the same weight), an even- $\# \text{Rac}$ state, so it rides the single cover (2π) — the paper does *not* claim gravity rides 4π . Moreover, since $(-1)^{\#\text{Rac}} = (-1)^F$ on two-singleton composites (Section 7’s premise; open per K1653), the Sakharov induced $1/G$ — a one-loop boson/fermion-signed sum — is a supertrace graded by $\# \text{Rac} \bmod 2$; we identify the *grading*, not a value for G . Disclaimer (not SUSY): the even- $\# \text{Rac}$ spin-2 ($\Delta=5$) and odd- $\# \text{Rac}$ spin-3/2 ($\Delta=9/2$) slots differ in weight by $1/2$ — a supercharge’s weight — but this is *not* supersymmetry: two towers are not an operator Q , and asserting one would collide with BST’s Five-Absence predictions. It is a graded *slot*, not a superpartner.

11b. Time and the forced object [Open]

The odd dimension $n_C = 5$ is the common source of the *constituent-level* half-integer structure throughout BST ($\sqrt{\pi}$, the Cabibbo $1/\sqrt{20}$, which Rac carries the half-odd weight). It forces *odd*, not 5; a principle forcing 5 specifically is open. [*Open.*]

12. What this paper does not claim (stated plainly — the paper’s strength)

- Physical time is not periodic or recurrent. The physical flow is a one-way arrow (Section 1); the degree-2 *circle* is the imaginary-time (Wick) face and a selection rule, not a loop in physical time. We never claim time comes back around.
- The tick’s value is Identified, not Derived (the exponent in $\approx 10^{-120}$ s is matched, not forced).
- The degree-2 cover is not a novel prediction — it is exactly fermion parity, in every dimension. We retract an earlier draft’s “the double cover is forced because $n_C = 5$ is odd — the paper’s core result”: on physical particles $\exp(2\pi i J) = (-1)^F$ dimension-independently (the constituent d -dependence cancels), so it is the geometric restatement of spin-statistics, a consistency the

theory reproduces. The odd dimension decides only *which constituent* is half-odd. We report the cover as this consistency; the paper’s substantive content is time-derived (Sections 1–4); the Section 11 causal-order material is a scaffold (the causal order is derived; the spectral = causal *weighting* is open). We also retract “only the Higgs rides 4π ” (it treated a bare singleton as a particle — fails the parity/spin filter, which also closes the old Rac=Higgs-vs-muon contradiction, both horns at once).

- The 4π is not observable by interferometry — it is a selection rule on admissible states (Section 6), not a measurable fringe.
- Gravity rides the single cover (the graviton is even-#Rac, 2π) — we do not claim gravity rides 4π . The induced-gravity supertrace (Section 11a) is an identification of the *grading* ($\#Rac \bmod 2 = (-1)^F$), not a value for G ; and the graded spin-2 / spin-3/2 slot is not supersymmetry — a slot, not a superpartner.
- The Higgs substrate identity is now closed (from primary source — a Lorentz scalar carrying an internal (1,0) vector, on the 2π cover; Section 7). It never affected the derived time content, and the Higgs is a boson (2π) on any reading. (We still name no physical particle as a *bare* singleton: a boundary singleton with no bulk field is not a bulk particle merely because both are “scalar” — the Higgs is the two-constituent composite, not a bare Rac.)
- The identity is exact for the free-field singleton weights $\{3/2, 2\}$; it is not protected against anomalous dimensions (Section 7).
- The boundary $\mathbb{Z}_2$ of the Shilov boundary $\mathbb{S} = (S^4 \times S^1)/\mathbb{Z}_2$ carries its own distinct -1 (a CPT-membrane structure); it is a different object from the time double cover — both live on the S^1 , but we keep them apart and do not merge them, and neither the boundary $\mathbb{Z}_2$ nor any Aharonov–Bohm reading is used in this paper’s derivations.
- Whether the two Wick faces and the degree-2 cover are one deeper object is a *candidate*, held apart, not claimed.
- Commitment forces the *type* of world (type-IV, rank 2) and the whole structure of time; it takes a small number of integers ($n_C = 5$ among them) as inputs — the same bargain general relativity makes, held to the same standard. We never claim “commitment forces D_{IV}^5 .”